

Financial Transactions

Data Warehouse

Homework 3 — Big Data Analytics · Simone Di Bella

DATA SOURCES	3 heterogeneous sources: account statement, symbols, country metadata
SCHEMA	Star schema with 1 fact table and 4 dimension tables
GRAIN	One row in Fact_Transactions represents one valid transaction
DASHBOARD	Time Analysis and Country Analysis pages

1 Introduction

This project develops a small data warehouse for the analysis of financial transactions recorded in a 2024 investment account statement. Starting from three heterogeneous data sources, the project follows the complete analytical workflow required for decision support systems.

The core of the project is a **star schema** designed to support multidimensional analyses of transactions across different dimensions. Particular attention is devoted to data quality assessment, since the source datasets contain inconsistencies and missing information that must be addressed before constructing the dimensional model.

Finally, the warehouse is used for analytical queries in Python and for an interactive **Streamlit dashboard**, allowing users to explore transaction patterns more easily.

2 Data Warehouse Design

The proposed data warehouse models the business process of recording financial transactions from the account statement. The **grain** of the fact table is defined at transaction level: one row in *Fact_Transactions* represents one valid transaction from the account statement file.

The model follows a **star schema** structure, with *Fact_Transactions* at the center and four dimension tables: *Dim_Time*, *Dim_Symbol*, *Dim_Geography*, and *Dim_TransactionType*. The fact table stores foreign keys and the numerical measure unit (number of units per transaction). Transaction counts are derived by aggregation rather than stored explicitly.

Each dimension serves a distinct analytical role: time supports day/month/quarter/year analyses; geography links stocks to country, sub-region, and region; symbol captures company, industry, and sector; and transaction type distinguishes BUY, SELL, and dividend records. **Surrogate keys** are used throughout, ensuring the model is independent from source attributes and simplifying joins.

3 Data Quality Assessment and ETL

The ETL pipeline integrates the account statement, the symbols dataset, and the country metadata into a consistent dimensional model. Before constructing the star schema, the source data were inspected to identify missing values, inconsistent identifiers, invalid dates, and integration problems.

The account statement initially contained **2,745 rows**. A first quality check revealed 464 rows with missing values in all main attributes; these were removed, leaving 2,281 cleaned records. The original *IDTransaction*

field was not unique and could not serve as a primary key; a new surrogate key *transaction_key* was therefore created.

Further integration checks identified 18 symbols (affecting 212 rows) that were absent from the symbols dataset. Since these records could not be linked to company, sector, industry, or country information, they were excluded to preserve referential integrity. Two country name inconsistencies (Taiwan and Turkey) were also corrected before joining the geography data. The final fact table contains **2,069 valid transactions**.

ETL Step	Row Count
Initial account statement	2,745
Empty rows removed	464
Rows after initial cleaning	2,281
Rows excluded (unmapped symbols)	212
Final fact table	2,069

Table 1 — Main data quality and ETL outcomes.

4 Analytical Findings

Eight analytical queries were implemented on a view obtained by joining the fact table with all dimension tables. The results reveal that transaction activity is concentrated in the **first half of 2024**: Q1 alone accounts for 968 BUY and SELL transactions, followed by Q2 (522), Q3 (242), and Q4 (241).

Geographically, the **United States** leads by number of SELL transactions (389), followed by the United Kingdom and China. At the regional level, the Americas account for the largest number of units bought (37,026). **Technology** is the top sector for SELL transactions in the US and for units traded in Q3, while **Semiconductors** appear prominently in Q4 BUY transactions and in industry-region pair analysis.

Analysis	Top Result	Value
SELL transactions in US by sector	Technology	158
BUY transactions in Q4 by industry	Semiconductors	18
BUY and SELL transactions by quarter	Q1	968
SELL transactions by country	United States	389
Units bought by region	Americas	37,026
Units traded in Q3 by sector	Technology	4,461
Transactions in Q4 by sub-region	Northern America	130
Industry-region pairs by transactions	Semiconductors – Americas	134

Table 2 — Summary of analytical query results.

5 Interactive Dashboard

The final component is an interactive dashboard built with **Streamlit**, translating the dimensional model into a user-facing analytical tool organised around two complementary views.

Time Analysis. Focuses on the temporal evolution of BUY and SELL transactions. A date range filter allows users to select the period of interest. This page includes the transaction trend over time and ranked bar

charts for the most traded symbols, sectors, and industries.

Country Analysis. Focuses on the geography associated with traded stocks. Users can select a country and inspect the corresponding transaction trend during 2024, together with the top industries by BUY and SELL transactions. If the selected country has no associated transactions, the dashboard displays a warning message instead of producing empty or misleading charts.

The dashboard uses line charts for temporal trends and bar charts for cross-sectional comparisons. Small formatting choices — integer axes, maximum-value markers — improve readability without adding unnecessary complexity.

6 Conclusion

This project transforms raw financial transaction data into a structured workflow for analysis, combining data cleaning, dimensional modelling, analytical queries, and interactive visualisation. The star schema makes transaction data easier to interpret and reuse by organising information around clear analytical dimensions and supporting consistent aggregations.

The ETL process plays a central role in ensuring that the final warehouse contains only records that can be correctly mapped and interpreted. Together, the dimensional model and the Streamlit dashboard turn a simple account statement into a decision-support tool: analysts can move from detailed transaction records to higher-level patterns, identify concentrations in specific markets or sectors, and explore the data interactively.